Rethinking Fall Detection With Wi-Fi

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Abstract—The past decades have witnessed a surge in human fall detection with sensors, cameras, and wireless signals. Among them, Wi-Fi-based fall detection has been one of the most attractive solutions due to the ubiquitous and pervasive deployment of Wi-Fi infrastructures. However, these approaches are still difficult to be put into practical use. To push forward Wi-Fi-based fall detection for wide deployment, three major limitations concerning *environmental diversity*, *motion diversity*, and *user diversity* are required to be resolved. In this paper, we propose FallDar, a Wi-Fi-based deep learning-assisted fall detection system that outperforms state-of-the-art works on the three criteria simultaneously. First, to deal with environmental diversity, FallDar characterizes falls with the speed of the body, which is the most relevant and inherent feature of falling activities, making the system resilient to environmental changes. Second, to deal with motion diversity, FallDar simulates a large amount of fall data of various falling types with a DNN-based generative model. Training with these data, FallDar is endowed the capability of detecting more types of falls. Third, to deal with user diversity, FallDar proposes to incorporate the fall detection network with a user identification network. The network is designed to extract user-independent features, requiring no fall data from new users for system adjustment. We implement FallDar on commercial Wi-Fi devices and conduct experiments in home and office environments for six months. The evaluation results show that FallDar achieves a false alarm rate of 5.7% and a missed alarm rate of 3.4% across all factors, making a fundamental step towards ubiquitous fall detection with Wi-Fi.

Index Terms—Fall detection, channel state information (CSI), commodity Wi-Fi

1 Introduction

Human falls are a major cause of injury and morbidity among senior citizens, which result in more than 2.8 million injuries treated in emergency departments annually, including over 800,000 hospitalizations and more than 27,000 deaths [1]. To prevent death and injury, extensive research efforts have been devoted to indoor fall detection. Traditional approaches have exploited the sensing ability of cameras [2], [3], [4], wearable sensors [5], [6], [7], or dedicated radar hardware [8], [9] for fall detection. However, cameras have the risk of privacy leakage, wearable sensors are intrusive to users and radars are expensive for large scale deployment. The demand for a device-free, secure, and ubiquitous fall detection mechanism has triggered interests in sensing with commodity Wi-Fi.

Previous Wi-Fi-based fall detection systems, such as WiFall [10], RTFall [11], FallDeFi [12], and DeFall [13] have demonstrated the feasibility of using Channel State Information (CSI) from Wi-Fi devices for fall detection. These systems, although yielding promising performance in specific settings, have three intrinsic limitations that leave them vulnerable to practical challenges in deployment scenarios.

- *Environmental factors are not completely removed from signals.* The signal features used in existing works are either primitive signal features such as amplitude and phase of CSI [10], [11] or Doppler Frequency Spectrogram (DFS) [12] converted from CSI. However, due to the multipath effects of wireless signals, these raw features are prone to environmental changes. The measured CSI readings portray the combined signal propagation paths including those from the surrounding environments, and DFS profiles are highly correlated with the locations and orientations of the users. Using these features, a fall detection system will have low generalizability between environments and should be retrained whenever deployment settings change. So, the performance of the existing works suffers from *environmental diversity*.
- *Human fall movements are not properly modeled.* Existing works typically associate falls with bursts of changes in CSI readings or frequency components of DFS. However, the actual falling movements are far beyond that perception. Human falls are complex movements that involve the collaborative movements of the torso and limbs, each of which has different moving patterns during falling. For example, some people may hold on to surrounding furniture or lean against the wall as they fall and some may fall from chairs, all of which induce less significant signal fluctuations. The diverse falling patterns will be misclassified as those normal activities that exhibit fast movements (e.g., running, abrupt sitting-down, etc). So, the performance of the existing works suffers from *motion diversity*.

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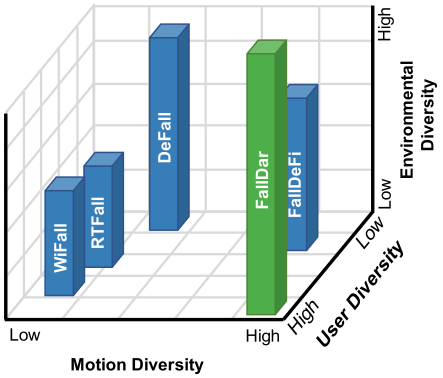


Fig. 1. A comparison of the state-of-the-art Wi-Fi-based fall detection systems. The three dimensions give a quantitative comparison between these systems. The Motion Diversity axis reflects the number of fall types reported in these systems. The User Diversity axis reflects the number of participants reported in these systems. The Environmental Diversity axis reflects the number of rooms evaluated in these systems.

- *Human discrepancies are not fully considered*. The existing works perform fall detection irrespective of human identity. However, as discussed in some works [14], [15] in the biological field, discrepancies exist in the falling patterns of different people, which are induced by diverse body characters such as weights, heights, and habitual motions. These discrepancies will perturb signals in different manners and pollute the performance when applying the systems to novel users. So, the performance of the existing works suffers from *user diversity*.

As a result, the pursuit of ubiquitous sensing necessitates an environment-independent, motion-irregular, and user-irrespective Wi-Fi-based fall detection system.

In this paper, we collaborate with a leading smart hardware manufacturing company to develop ***FallDar***, an easy-to-deploy ***Fall****detection ra****Dar*** based on commodity Wi-Fi routers. The ultimate goal of FallDar is to provide a low-cost, sensorless, and contactless fall detection system commercially available to the elderly that live on their own. Fig. 1 shows a qualitative comparison between FallDar and the state-of-the-art works. FallDar intends to achieve consistent performance across all three factors simultaneously. First, FallDar intends to have prominent generalizability between different deployment environments. Second, FallDar intends to retain robustness to human motion variations. Third, FallDar intends to be readily scalable between users. To support these features, FallDar has to overcome three technical challenges.

The first challenge is to achieve consistent performance across diverse environments. Based on our observations, the key to solve this problem is to extract features that are resilient to environmental variations. FallDar proposes to characterize fall movements with the moving *speed* of the human body, which conveys the most relevant information to human activities compared to CSI or DFS. To extract speed from CSI signals, some existing works [16], [17] have established the quantitative relationship between the speed of moving objects and the DFS profile extracted from CSI. However, they all assume that the Line-of-Sight (LOS) path between the Wi-Fi devices and the object dominates in the measured signal, which is unrealistic for typical household scenarios. To remedy this, we design a statistical approach for speed estimation under Non-Line-of-Sight (NLOS) and rich-scattering settings. Specifically, we regard each object in the indoor environment as a latent scatterer that would diffuse the signal away in all directions. By modeling the signals emitted by each scatterer as a Gaussian random variable, we derive the quantitative relationship between the moving speed of scatterers and the statistical features of the observed CSI. This method discards the specific signal propagation path and is resilient to environmental reflections.

The second challenge is to achieve consistent performance for different falling motions. The key to solving this problem is to build a large and representative training dataset that encompasses diverse types of fall movements. However, collecting fall data requires tremendous effort and even if this dataset is collected, there remain unseen fall types which the classifier may not be able to generalize to. To tackle this problem, we propose to augment the training dataset by generating synthetic data instances with a generative model based on Deep Neural Network (DNN). We first analyze the speed transformations of various types of falls and identify that the diversity of falls comes from the protective movements of limbs. Thus, we simulate perturbations induced by the limbs' motion with the generative model and produce other new types of fall instances. With the cooperation of collected and synthetic fall data, FallDar is endowed with the capability of detecting more types of fall activities.

The third challenge is to achieve consistent performance across different users. Our key insight is to remove user identity from the features used for fall detection. To achieve this purpose, FallDar proposes to incorporate a human identification task into the fall detection task. Specifically, we build a DNN-based network that can output a fall indicator and human identity simultaneously. Different from traditional classifiers, we deliberately force the network to output false human identity, which in turn fraud the latent features of the network to be user-independent. The classifier can be trained with widely available walking data of the target people, which imposes a minor burden on model adaptation. Therefore, FallDar is immune to user diversity.

Putting it all together, FallDar is implemented on several commercial Wi-Fi routers and evaluated with a voluntary family across 6 months. The participants consist of two elderly members of the same age of 50 and three young members between the ages of 20 and 25. We collect 3,342 data instances after 6 months of continuous monitoring, during which 21 types of normal activities and 24 sub-variations of fall movements were observed. Evaluation results have demonstrated that FallDar can achieve an average false alarm rate of 5.7% and a missed alarm rate of 3.4% across all factors (environments, fall types, and users) in typical home scenarios, outperforming the state-of-the-art works, making it the most promissing Wi-Fi-based fall detection system for commercial use.

In a nutshell, we make three main contributions to push the limit of Wi-Fi-based fall detection:

- We resolve the motion diversity and user diversity problem in the fall detection system by disentangling the fall-specific information from the falling speed

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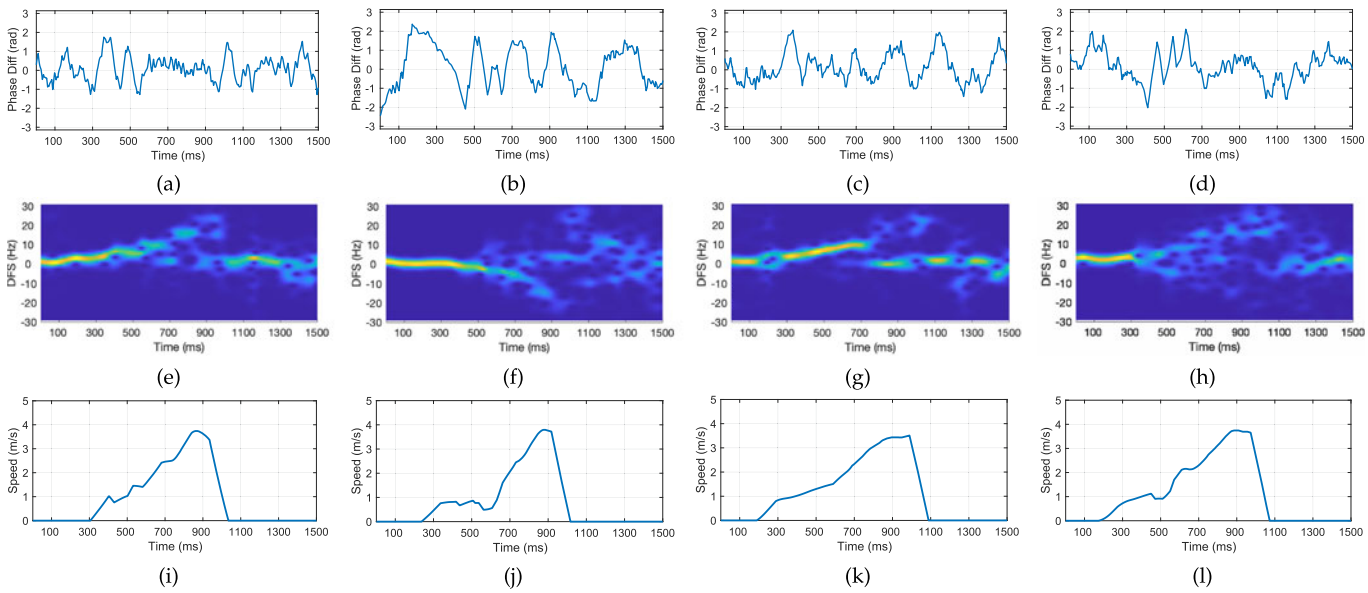


Fig. 2. Phase difference, DFS, and speed of torso features of falls in different environment settings. (a)(e)(i) Location #1, orientation #1. (b)(f)(j) Location #1, orientation #2. (c)(g)(k) Location #2, orientation #1. (d)(h)(l) Transmitter and receiver are placed in different rooms.

pattern with signal processing and deep learning techniques respectively.

- We build a rigorous fall detection system FallDar on commodity Wi-Fi devices, which can be easily generalized to new users and detect diverse falling patterns in various environments, making the large scale deployment of fall detection systems in real business scenarios readily available.
- We conduct extensive experiments in authentic household scenarios, which demonstrate a consistent performance of FallDar across diverse factors and confirm its prominent potential for ubiquitous use.

The rest of the paper is organized as follows. We introduce the observations and motivations of FallDar in Section 2, system overview in Section 3, technical details in Sections 4, 5, and 6. We present evaluation in Section 7, review the literature in Section 8, and conclude this paper in Section 10.

2 Observations and Motivations

To exploit the gap between research and application of Wi-Fi-based fall detection systems, we conduct extensive field experiments to study to what extent different factors affect fall detection systems. Specifically, we deploy commercial Wi-Fi routers in a family house to collect CSI measurements. Several observations are made based on the results.

2.1 Impact of Environmental Diversity

In FallDar, *environmental diversity* includes both the static surroundings (furniture, walls, ceils, etc) and the relative locations and orientations of the user to Wi-Fi devices. In practical deployment scenarios, the fall detection system can be deployed in diverse families and houses with different layouts. More importantly, the moving locations and orientations of the users are not fixed, making the robust detection much more challenging. Hence, it is necessary to study the impact of environmental diversities on the sensing system.

To reveal the correlation between signal features and environmental factors, we set up four data collection settings in a family house. In the first three settings, a pair of Wi-Fi devices are placed in the sitting room with a distance of 3 meters. A user performs fall movements (with protective equipment) at different locations and orientations. In the last setting, a pair of devices are put in two different rooms with LOS occluded by the wall. After collecting CSI measurements, we subtract the CSI phase between two adjacent antennas to mitigate the random phase noise [11]. Besides, we perform Short-time Fourier Transform (STFT) to extract the DFS, which was also used for fall detection in [12]. We extract human moving speed with the speed estimation algorithm to be introduced in Section 4.

Figs. 2a, 2b, 2c, 2d, 2e, 2f, 2g, 2h, 2i, 2j, 2k, and 2l demonstrate the phase difference, DFS, and speed during human falls. Four observations can be made according to the results. (1) The CSI phase difference is highly sensitive to the environment. For both the dynamic and static period during falling, the phase difference changes drastically with no conspicuous pattern, making it an undesirable motion indicator for fall detection. (2) When the user falls at the same location (location #1) but to different orientations (orientation #1 and #2), the DFS profiles demonstrate different frequency shifts (Figs. 2e and 2f even with contrary signs. When the user falls to the same orientation (orientation #1) but at different locations (location #1 and #2), the DFS profiles reveal different degrees of frequency shift (Figs. 2e and 2g). (3) When the signal is occluded by walls, frequency shifts in the DFS profile become weak and faint (Fig. 2h). (4) The speed feature demonstrates a consistent pattern across diverse environmental factors.

From the above observations, we can conclude that traditional signal features including CSI phase difference and DFS are prone to environmental changes, making them suboptimal for fall motion characterization. To build a fully environment-independent fall detection system, we try to

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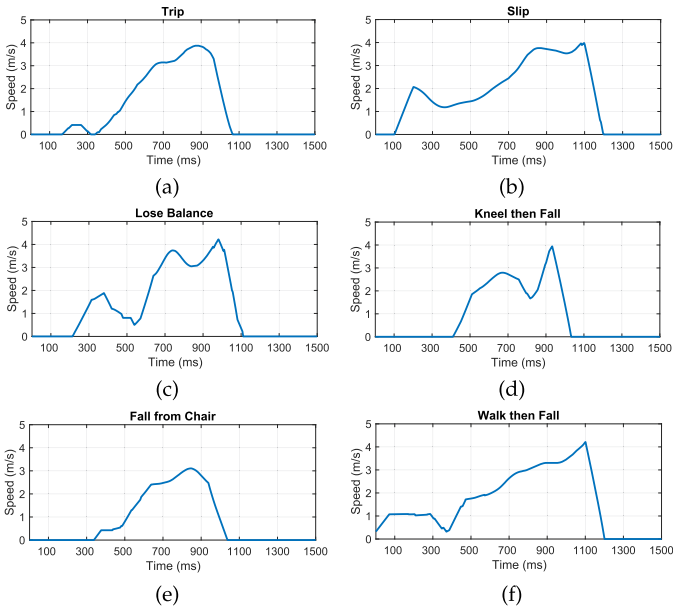


Fig. 3. Speed transformation of different falling types. (a) Trip. (b) Slip. (c) Lose Balance. (d) Kneel then fall. (e) Fall from chair. (f) Walk then fall.

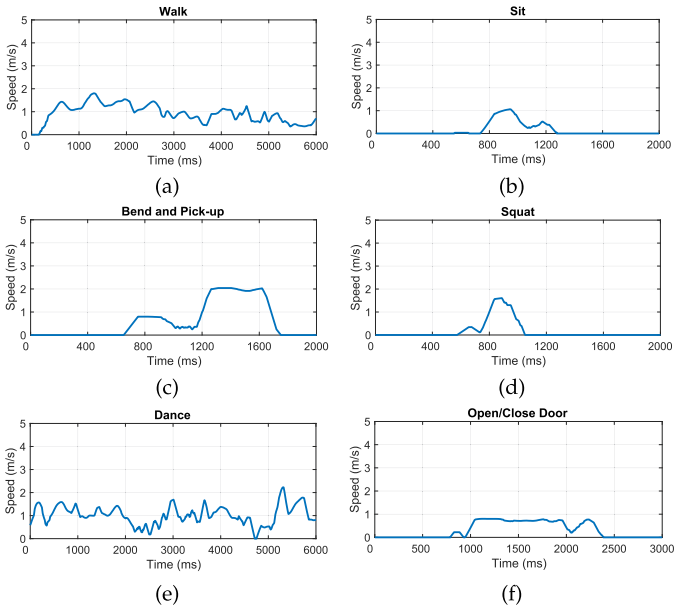


Fig. 4. Speed transformation of different normal motions. (a) Walk. (b) Sit. (c) Bend and pick-up. (d) Squat. (e) Dance. (f) Open and close door.

extract human torso speed from raw CSI signals as a motion indicator, and Section 4 is designed for this purpose.

2.2 Impact of Motion Diversity

Human falls are complex motions that involve the collaborative movements of the full body. In practice, the users can fall in different manners, making the fall detection task much complicated. Existing Wi-Fi-based fall detection systems ([10], [11], [12]) model human falls as *sudden* torso movements that incur bursts of changes in the signal measurements. However, in this paper, we ask the following question: Does the word *sudden* suffice to portray human falls?

To have a full understanding of human falls, we investigate how the torso moves when the human perform different fall and normal activities. We ask a user to imitate six falling motions including trip, slip, lose balance, kneel then fall, fall from chair, and walk then fall. During the experiment, we measure CSI with Wi-Fi and extract human moving speed with the proposed speed estimation algorithm. As a comparison study, we also ask the user to do some normal activities including walk, sit, bend and pick-up, squat, dance, open and close the door.

Figs. 3a, 3b, 3c, 3d, 3e, and 3f demonstrate the speed transformation of the six fall motions, and Figs. 4a, 4b, 4c, 4d, 4e, and 4f demonstrate the speed transformation of the six normal motions. As can be seen, even though fall motions have larger speeds and accelerations, the discrepancies between different falling types are non-trivial. This observation motivates us to believe that by modeling fall motion in a fine-grained manner, we can have an in-depth understanding of human motion, which will potentially help us to achieve better performance in fall detection.

To dive deep into the impact of limbs motion, we evaluate the falling speed with regard to different moving patterns of limbs. We first ask a volunteer to fall from a standing position without moving his legs and arms. Then, we ask the volunteer to move his arms freely when falling but keep the legs fixed. At last, we ask the volunteer to take a step forward and move his arms freely when falling. The falling speeds of the above three patterns are shown in Fig. 5. Without limbs motion, the falling speed has the largest peak value and the shortest duration. The acceleration stage is more steady and smooth than the other patterns. When the hands are free during the fall, the volunteer will reach his hands to seek balance, which will cause the falling speed to decelerate. The peak value of speed is smaller and the falling duration is longer in this case. When the limbs all move freely, the acceleration becomes more mild, revealing a smaller peak value and a longer duration. The above evaluation results reveal that the falling patterns are greatly affected by the motion diversity of limbs. In Section 5, we try to leverage the motion diversity to strengthen the system performance.

2.3 Impact of User Diversity

A commercially valuable fall detection system should achieve consistent performance for different users. That makes us think if user diversity is an important factor to consider when building a fall detection system.

To validate the impact of user diversity, we compare the falling patterns of different users. We invite five users to perform trip falls. Similar to the above experiments, the human moving speed is extracted from the CSI readings captured by a pair of Wi-Fi devices. Besides, we also use a dummy to imitate the falling of humans (details can be found in Section 5.1). We tie a smartphone on the waist of the dummy to help measure the ground-truth speed.

Figs. 6a, 6b, 6c, 6d, 6e, and 6f demonstrate the speed transformation of the dummy and the five users. Several observations can be made. First, compared with the dummy that has no consciousness, the falling speed of human have more fluctuations due to the protective movements during falling. Second, the user with low heights (user #5 and #6) have smaller peak speeds during falling. Besides, the duration of falls varies between users.

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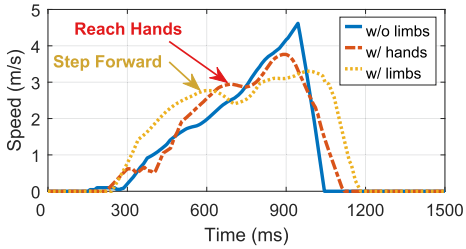


Fig. 5. Impact of limbs motion on falling speed.

The above observations reveal that discrepancies exist in the falling pattern of different users. To further evaluate how would these discrepancies affect fall detection accuracy, we implement a prototype model based on an SVM classifier. The model takes as input the DFS profile of both fall and normal instances. We train the model with the data from one user and tested it with the data from one of the other four users. The detection accuracy is shown in Fig. 7. As can be seen, without training samples from the testing user, the detection accuracy would tumble by a significant rate. Hence, a fall detection system demands to take the diversity of users into account. Section 5 is designed to mitigate the impact of user diversity on the detection system.

3 System Overview

FallDar inputs raw CSI streams measured by a pair of Wi-Fi devices and outputs whether fall movements are detected. It is composed of three main modules: Data Collection module, Data Generation module, and Fall Detection module. Fig. 8 presents the detailed system workflow.

In the Data Collection module, CSI measurements of fall and normal movements are processed by a speed estimation algorithm to extract the actual moving speed of the human body, which is a physically plausible and environmentally independent motion feature for fall detection, making it more robust to environmental diversity. Upon obtaining the speed sequence of fall activities, the Data Generation module

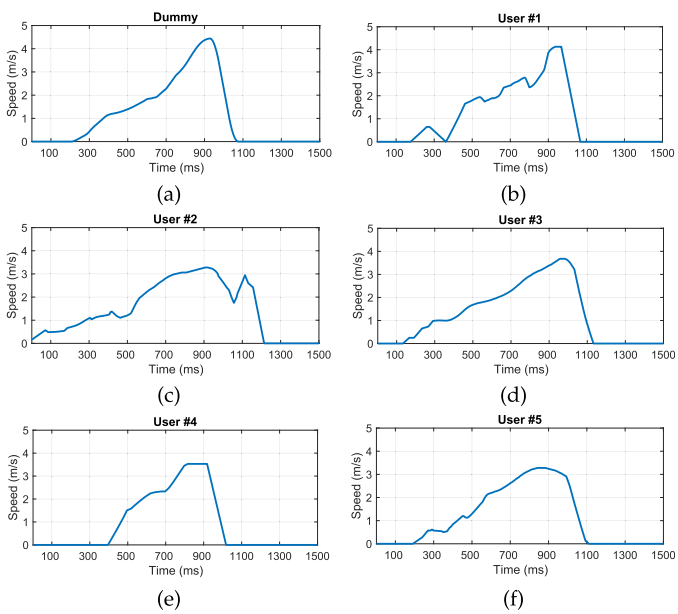


Fig. 6. Speed transformation of falls from the dummy and user #1 to #5.

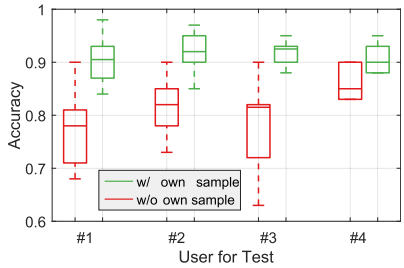


Fig. 7. Fall detection accuracy across users.

leverages a generative DNN network to synthesize human limbs' movements, which is aimed to augment fall samples in training datasets and generates more fall types that are not included in the collected datasets. With the augmented training sets, FallDar is capable of detecting more types of fall movements, making it more robust to motion diversity. The collected fall and normal samples, as well as the synthetic fall samples, are passed to the Fall Detection module. This module is based on an adversarial DNN network that can recognize fall movements and users' identities simultaneously. However, we deliberately encourage the network to misidentify the users, forcing it to extract user-independent latent features. By doing so, the fall detection results are more robust to user diversity.

4 Fall Speed Estimation

Existing Wi-Fi based fall detection systems extract features from CSI or DFS as fall representations, which are prone to environment changes. FallDar proposes to characterize fall motion with speed of human torso estimated with Channel State Information (CSI).

CSI portrayed by commodity Wi-Fi can be modeled as:

$$H(f,t) = \sum_{l=1}^{L} \alpha_l(f,t) e^{-j2\pi \int_{-\infty}^{t} f_{D_l}(f,\tau)d\tau}, \quad (1)$$

where L is the total number of reflection paths, α_l is the amplitude attenuation of path l, f_{D_l} is the DFS induced by object movements on path l.

Some ray-tracing-based works [16], [17] geometrically trace the signal propagation paths and derive the relationship between DFS and object moving speed. However, they have limitations in two aspects. First, DFS is correlated with the relative location and orientation of moving objects

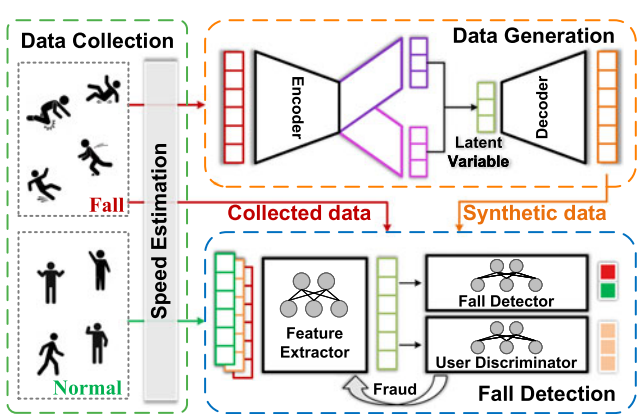


Fig. 8. Workflow of FallDar.

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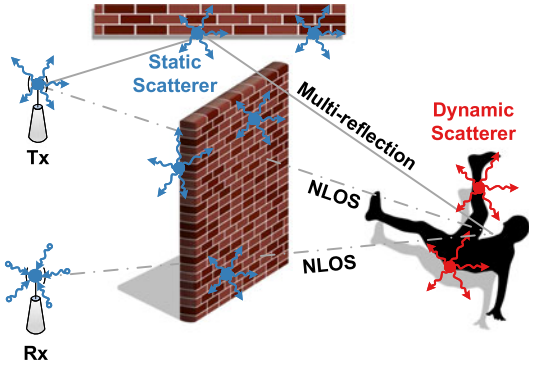


Fig. 9. Wi-Fi signals in rich-scattering environment.

related to the Wi-Fi link. Specifically, only the radial speed projected on the normal direction of the ellipse with Tx and Rx as foci could produce DFS. In our previous work Widar3.0 [18], we have already demonstrated the feasibility of speed estimation in an environment-independent manner. However, Widar3.0 is based on the assumption that there is a direct path between Wi-Fi devices and the target user, which is reasonable for gesture recognition tasks. The condition can hardly be met in fall detection tasks because the user can fall at any indoor location. Hence, a novel speed estimation algorithm that can work in NLOS scenarios is necessary. Second, the signal propagation paths in typical indoor environments can hardly be traced due to the complex multipath effect and occlusions. As a straightforward example in Fig. 9, a user falls behind a wall with no LOS path to the Wi-Fi transceivers. The received CSI is dominated by the multi-reflection components with potentially several times reflections on the surrounding objects, which limits the application prospects of the ray-tracing-based speed estimation methods.

To enable speed estimation in multipath-rich environments, we design a statistical approach inspired by [19], [20]. As is shown in Fig. 9, we treat all the objects in indoor environments as *scatterers* that diffuse the signals in all directions. The CSI observed by the receiver is added up with the portions contributed by all of the static (like furniture, walls, etc) and dynamic (like arms, legs, etc) scatterers. Intuitively, each scatter is deemed as a "virtual Tx". Such modeling is tenable in typical indoor scenarios, where the rooms are crowded with furniture and signals could propagate in all directions. With this model, we dismiss the specific signal propagation path and intend to statistically investigate the correlation between the observed CSI and the moving speed.

We decompose the observed CSI into the portions contributed by individual scatterers:

$$H(f,t) = \sum_{i \in \Omega_s(t)} H_i(f,t) + \sum_{j \in \Omega_d(t)} H_j(f,t), \tag{2}$$

where $\Omega_s(t)$ and $\Omega_d(t)$ are the sets of static and dynamic scatterers, $H_i(f,t)$ is the portion of observed CSI contributed by i^{th} scatterer. For each scatterer, the diffused components undergo anonymous propagation processes and eventually add up at the receiver. If setting a three-dimensional coordinate with its origin at the ith scatterer and its Z-axis align with the scatterer's moving direction, the representation of $H_i(f,t)$ is further decomposed as follows [19]:

$$H_i(f,t) = \int_0^{2\pi} \int_0^{\pi} h_i(\alpha,\beta,f,t) exp(-jkv_i cos(\alpha)t) d\alpha d\beta, \tag{3}$$

where $k = 2\pi/\lambda$ and λ is the signal wavelength, v_i is the speed of ith scatterer, α and β are the pitch and yaw angles, $exp(-jkv_i cos(\alpha)t)$ represents the DFS of the signal on direction (α,β). $h_i(\alpha,\beta,f,t)$ is the portion of observed CSI contributed by i^{th} scatterer on direction (α,β). Inherited from the physical properties of EM waves [21], $h_i(\alpha,\beta,f,t)$ can be viewed as a circularly-symmetric Gaussian random variable. For $\forall i,j \in \Omega_d$ and $(\alpha_1,\beta_1) \neq (\alpha_2,\beta_2)$, $h_i(\alpha_1,\beta_1,f,t)$ is independent of $h_j(\alpha_2,\beta_2,f,t)$.

Based on Equations (2) and (3), we are able to derive the statistical property by analyzing the autocorrelation function (ACF) of $H(f,t)$ [21]:

$$\begin{aligned}\rho_H(f,\tau) &\triangleq \frac{Cov[H(f,t),H(f,t+\tau)]}{Cov[H(f,t),H(f,t)]} \\ &= \frac{\sum_{i\in\Omega_d} \sigma_i^2(f) sinc(kv_i\tau)}{\sum_{i\in\Omega_d} \sigma_i^2(f)} \approx sinc(kv\tau),\end{aligned} \tag{4}$$

where $Cov[\cdot,\cdot]$ is the covariance between two random variables, $\sigma_i^2(f)$ is the variance of $h_i(\alpha,\beta,f,t)$, $sinc(kv\tau) = \frac{sin(kv\tau)}{kv\tau}$, and τ is the time lag. For human fall activity, we can assume that the dynamic scaterers are mostly on the torso and have similar speed v. The rationality behind this assumption is that the torso contributes most of the strong scatters, which have similar speeds and dominate those from the limbs [20]. Even though we can discard the scatterers on the limbs, the reciprocal influence on the human torso is non-trivial. In the next section (Section 5), we will discuss how will the protective movements of limbs affect the speed of the torso during falls. Thus, Equation (4) is approximated with a much simpler form: $\rho_H(f,\tau) \approx sinc(kv\tau)$. Using this formulation, we have quantitatively established the relationship between the speed of human and the ACF of CSI. In practice, the speed v is extracted by matching the first peak of the $sinc(x)$ function and the first peak of ACF:

$$v = \frac{x_0}{k\tau_0} = \frac{x_0\lambda}{2\pi\tau_0}, \tag{5}$$

where x_0 is the constant value representing the location of the first peak of $sinc(x)$ function, τ_0 is the time lag corresponding to the first peak in the ACF.

As a straightforward example, we invite a volunteer to walk and fall in a heavily furnished room and extract the speed with the CSI collected from a Wi-Fi link located in another room. As is shown in Fig. 10, even with NLOS occlusions and multipath pollution, the walking speed and fall speed are estimated consistently, demonstrating its robustness to environmental diversity.

Comparison Between DFS and Speed. The statistical signal model can produce the speed of the target irrespective of its moving directions, while the DFS only captures the radial speed projected on the normal direction of the ellipse with Tx and Rx as foci. To compare DFS and speed for different moving directions, we design two experiments. In the first experiment, we let a volunteer free fall in the normal and tangential direction of the Wi-Fi link. The DFS and speed

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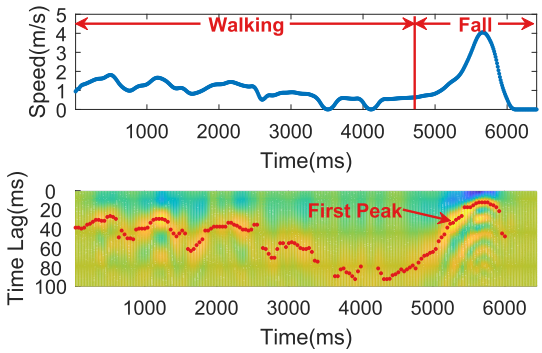


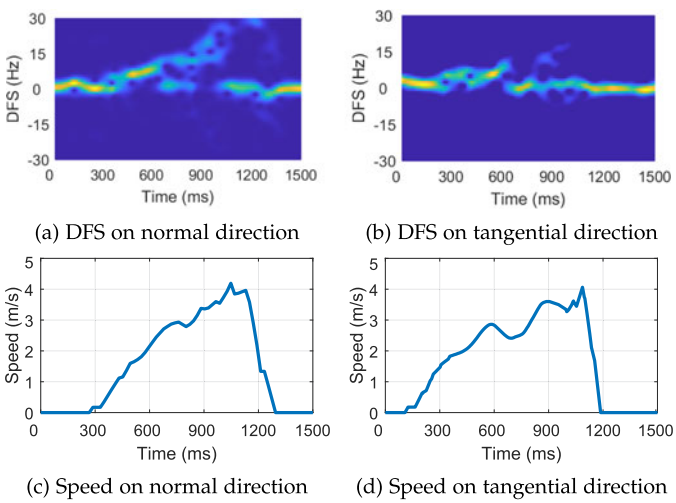
Fig. 10. The estimated speed and the ACF of CSI signals.

extracted from the statistical model are shown in Fig. 11. Clearly, while falling in the normal direction induces drastic doppler frequency changes, falling in the tangential direction only creates minor doppler frequencies, which will be mistaken for other activities. On the contrary, the statistical model produces consistent speed in both cases. In the second experiment, we let a volunteer walk on an L-shaped track. Specifically, the volunteer first walks in the normal direction and then walks in the tangential direction, and returns the same way. The DFS and speed are shown in Fig. 12. It is observed that when the target moves in the tangential direction, the DFS is less observable than that from the normal direction. However, the calculated speed is more robust to the moving directions. Hence, the fall speed estimation algorithm is a better choice to tackle the environment diversity problem.

In the next sections, we will discuss the methods to detect fall motion with the estimated speed information.

5 Fall Data Augmentation

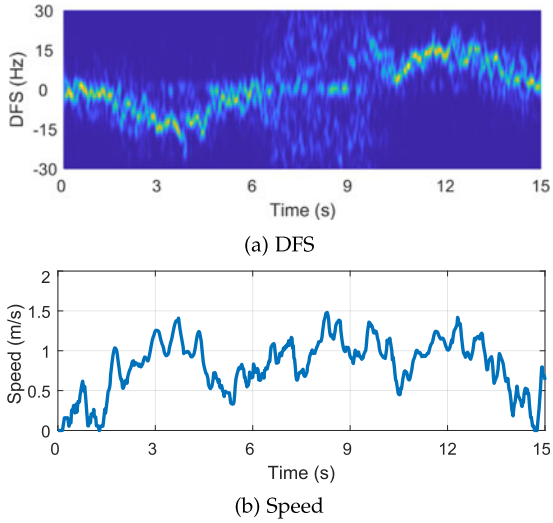
Existing fall detection works are based on the presumption that fall motion induces faster torso speed than other daily activities and could be identified by analyzing the abrupt changes in signals. However, not all falls have to be fast. In practice, people may hold on to surrounding furniture or lean against the wall as they fall and some elderly may even



(a) DFS on normal direction (b) DFS on tangential direction

(c) Speed on normal direction (d) Speed on tangential direction

Fig. 11. Comparison between DFS and speed for different falling directions.



(a) DFS

(b) Speed

Fig. 12. Comparison between DFS and speed for walking on a non-linear track.

fall from the chair. In such cases, the moving speed of the torso is not significant, hence diluting the basic foundations of the past works. Worse still, some daily activities such as sitting down, run or dance also exhibit fast motion and would be misclassified as fall.

To deal with the complexity of body motion, FallDar demands a large and representative training dataset with a variety of fall types. However, the problem of enriching the training set is twofold. First, human fall activity is rarely seen in daily life. Collecting large datasets with an abundant number of falls is time-consuming and labor-intensive. Second, there are enormous types of fall motion such as walk-then-fall, sit-then-fall, trip, slip, etc. Even if we manage to collect datasets with plentiful fall samples, there would remain unseen fall types to which the classifier may not be able to generalize. The key observation to solve this issue is that the falling speed of the torso follows a unique pattern with two phases, i.e., acceleration and deceleration, corresponding to the falling and striking period of falls. The motion diversity is caused by the random limbs movements and their reciprocal influence on the torso. Based on this observation, it is straightforward to find out this unique pattern and add some perturbations to simulate falls. In this section, FallDar intends to solve these issues by leveraging a stochastic generative model based on DNN. This model can generate a large amount of realistic synthetic data from a small number of real-world measurements.

Below, we describe how FallDar models human fall activity and generates diverse fall samples with VAE model.

5.1 The Kernel Speed of Fall

Human fall is a complex body motion involving the movements of the torso and limbs. Biomechanics research [14] reveals that instinctive movements of the limbs exist for preventing injury during a fall. For example, upper extremities would sway, legs would step towards the falling direction and even knees would bend to contact the ground to seek balance when falling. Such protective movements, though less perceptible, could affect the fall speed and lead to motion diversity of falls. This is similar to shooting a basketball, where a player jumps to throw a ball forwards and his

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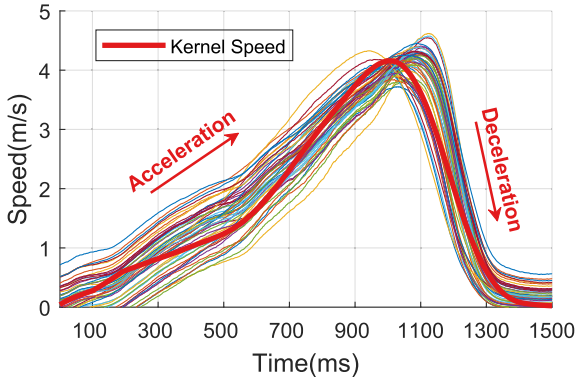


Fig. 13. Falling pattern of dummy (Kernel Speed) and human.

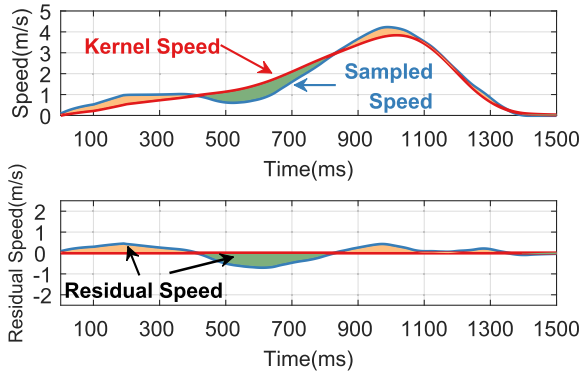


Fig. 14. Kernel speed and residual speed of fall.

torso naturally leans backward. Without the instinctive movements of the limbs, the whole body will be dragged by gravity and fall freely to the ground in an uncontrolled manner. Without loss of generality, if the initial posture and body structure are fixed, we can assume that this uncontrolled manner of falls has an identical speed pattern. In another word, the diverse falling patterns are perturbed versions of a fixed pattern. In FallDar, we model human falls as two additive components, i.e., the *Kernel Pattern* and the *Residual Pattern*. The Kernel Pattern corresponds to the speed of the human body for uncontrolled falling, and the Residual Pattern corresponds to the speed perturbations induced by limbs. We define the speed of the human body as the moving speed of the upper torso. In the following parts, we discuss the procedures to capture the Kernel Pattern and the algorithm to simulate the randomness in the Residual Pattern.

Based on the definition of Kernel Pattern, it is a reasonable way to capture this pattern from a dummy that has no active moving capability. The dummy used in FallDar has the same height as an ordinary person and is covered with latex rubber, a material that is similar to human skin. Note that the body structure of the dummy has slight differences with humans and we will discuss this phenomenon in Section 6. We let the dummy free fall from standing posture and estimate its falling speed with the algorithms in Section 4. Multiple experimental results are averaged to obtain the *Kernel Speed*. Fig. 13 demonstrates the *Kernel Speed* and fifty samples of human falls. As can be seen, a fall activity is composed of two main stages. The acceleration stage reveals an increase of speed from 0 to about 4 m/s, lasting for about 1000 milliseconds. The deceleration stage reveals a tumble of speed towards 0 m/s, lasting for about 300 milliseconds. The fall patterns from real persons exhibit diversity due to the protective movements of limbs, but all comply with the *Kernel Speed*.

After representing the core characteristics of fall with *Kernel Speed*, we intend to simulate the perturbations induced by the complex motion of the limbs. Specifically, we subtract the *Kernel Speed* from the fall speed sampled from real persons and obtain *Residual Speed* that is independent of the fall movements. An example of the two speed patterns is shown in Fig. 14. *Residual Speed* embodies the relative motion of limbs to the torso, which contributes to the diversity of various fall types With *Residual Speed* as input, the generative model is trained to learn the latent representation of limbs' motion during falls and generalize to more complex fall movements.

5.2 The Generative Model

Among various kinds of generative models [22], [23], FallDar employs Variational Autoencoder (VAE) [23] because it can generate large amounts of samples from a small input dataset. The core idea underpinning VAE is that the input data can be represented with a normal distribution in much lower-dimensional space. VAE intends to train an Encoder network that encodes the input data into this latent space and reconstruct it from this space with a Decoder network. Once trained, VAE can draw random samples from this space and generate new data with the Decoder network. In the context of our fall detection task, the fall diversity is induced by the kinetic movements of limbs, which is characterized in the *Residual Speed* features. VAE takes as input *Residual Speed*, captures the underlying distribution with network parameters, and generates synthetic data for unseen limbs' movements. Such movements, when combined with *Kernel Speed*, would derive novel fall types.

The VAE network used in FallDar is illustrated in Fig. 15. It is composed of two main components: **Encoder** and **Decoder** networks. The input x of Encoder is a 1-D vector with a length of 1,500, representing 1,500 snapshots of *Residual Speed* within 1.5 seconds. Two Fully-connected (FC) layers activated by rectified linear unit (ReLU) are cascaded to extract high-level features from x. Another two FC layers are adopted to extract the mean (μ) and variance (σ) of the latent vector (Z). To enable backpropagation in the latent space, VAE employs a trick called *reparameterization*. Specifically, vectors randomly

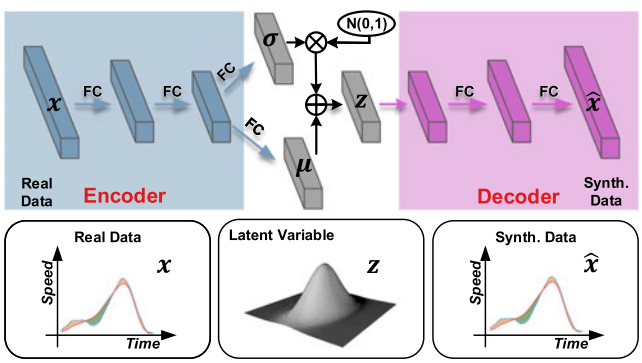


Fig. 15. FallDar's variational autoencoder.

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sampled from the standard normal distribution are multiplied with σ and added with μ to generate a latent vector Z, and thus Z is regulated to comply with a normal distribution. The Decoder network is composed of two FC layers activated by ReLU and Sigmoid to map Z into synthetic data $\hat{x}$. The learning objective of VAE is to minimize the reconstruction loss between x and $\hat{x}$ and regulate Z into a standard normal distribution ($N(0,1)$). Formally, the loss function is defined as:

$$\mathfrak{Loss} = \mathcal{KL}[N(\mu,\sigma), N(0,1)] + C\|x-\hat{x}\|^2, \tag{6}$$

where $\mathcal{KL}(\cdot,\cdot)$ is the Kullback-Leibler divergence between two distributions, $\|\cdot\|^2$ is the euclidean distance between two vectors, and C is the constant weight.

After training the VAE with a small dataset of fall samples collected on-site, FallDar randomly draws samples from the standard normal distribution and passes them into Decoder to generate synthetic data. These data, after incorporating with *Kernel Speed*, can be directly used as the training set of the fall detection classifier. Theoretically, we can draw an endless number of samples from the model. Therefore, using the augmented datasets in training, FallDar can generalize its classifier to detect unseen fall types and increase its robustness to motion diversity of humans.

6 Fall Detection

In some of the application scenarios, a practical fall detection system is supposed to detect fall motion while identifying the user's identity, which can accommodate the post-fall assistant strategy for individuals. However, based on our discussions in Section 2, the distributions of fall speed for different users have significant discrepancies, which will potentially create bias in pattern recognition and reduce the system performance. In FallDar, to maximize the performance, we attempt to mitigate the impact of user diversity by removing users' identity information from the wireless signals. Arguably, even though FallDar discards users' identity, the system will still be valuable in most of the fall detection scenarios. This is because FallDar targets the users that live on their own, whose identity is already known before deployment and there is no need to discriminate from other users. To enable fully user-independent fall detection, a challenging yet practical problem should be resolved: a fall detection pipeline that is trained on a predefined user set can be directly applied to other users without extra training efforts. This problem is especially challenging as we need to remove the user identity information from the signals while preserving the fall features before passing them into the fall classifier. As far as web know, none of the existing fall detection works can perform such procedures.

To remedy the user identity problem, FallDar intends to incorporate a user identification task into the fall detection task. Our inspiration is that some state-of-the-art wireless sensing works [20], [24], [25] have demonstrated the potential of human identification based on the gait information with commodity Wi-Fi devices. In one's everyday life, his/her walking data are widely available, which contains abundant identity components. FallDar uses these data to remove user-specific features and preserves the sanitized features for fall detection. In FallDar, we leverage the adversarial learning [26] technique based on DNN to achieve this purpose. Fig. 16 provides the DNN architecture. We design a User Discriminator to recognize the user by whom the activities are performed. This network and Fall Detector share the same input features, which are extracted by a Feature Extractor network. We confuse the discriminator and force the Feature Extractor to generate features of activities independent of users.

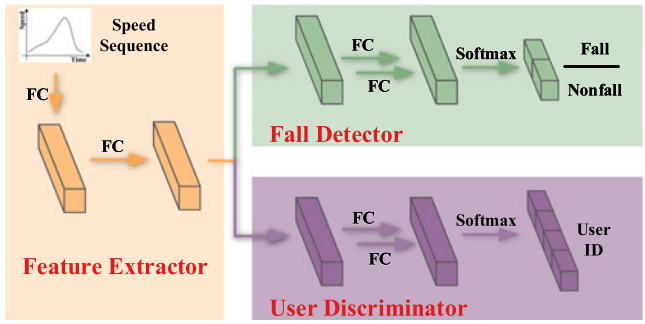


Fig. 16. FallDar's adversarial learning-based fall detection network.

Let $\mathcal{S}$ be the input datasets. During training, each sample $s \in \mathcal{S}$ is a one-dimensional vector with a time length of T. The fall samples are generated by the VAE model in Section 5.2 and the normal samples are from both the pre-collected dataset and the custom samples of the target users. s is labeled with fall or normal and user identity. During testing, FallDar applies sliding windows on the estimated speed stream with a window length of T and a stride length of $T/2$. Each window is evaluated with the detection network. We empirically set $T = 1.5$ seconds. Feature Extractor is meant to extract latent features from the input speed sequences. FallDar employs two FC layers to map the input $\mathcal{S}$ into a latent space:

$$F = f(f(\mathcal{S},\theta_1),\theta_2), \tag{7}$$

where θ_1, θ_2 are the parameters, F is the extracted feature. The FCs are activated by ReLU to introduce non-linearity. Dropout mechanism is adopted before each FC to avoid overfitting. By mapping the input into the latent space, we remove the noise in the speed sequence and capture the continuity and the correlation over time. F is further fed into Fall Detector for fall state inference. In the Fall Detector network, two FC layers and a Softmax layer are applied to obtain the fall indicator:

$$O_{fall} = softmax(f(f(F,\theta_3),\theta_4),\theta_5), \tag{8}$$

where $\theta_3, \theta_4, \theta_5$ are the parameters. The Softmax layer has two output units and O_{fall} is a one-hot encoding vector indicating fall or normal. As is discussed earlier, $\mathcal{S}$ contains users' identity that could deteriorate the detection accuracy, which is also the case for the latent feature F. We pass F into User Discriminator to identify the users' labels:

$$O_{user} = softmax(f(f(F,\theta_6),\theta_7),\theta_8), \tag{9}$$

where $\theta_6, \theta_7, \theta_8$ are the parameters. For a dataset of N users, the Softmax has N output units, and O_{user} is a one-hot encoding vector indicating the identities of users. The goal of User Discriminator is to force the latent feature F to be

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independent of the users' identity to boost the performance of Fall Detector. Hence we have to maximize the loss in User Discriminator while minimum the loss in Fall Detector. The final loss function for the DNN network is:

$$\mathfrak{Loss} = L_{fall} - \alpha L_{user}, \tag{10}$$

where α is a constant, L_{fall} is the binary cross-entropy between O_{fall} and the label of fall, L_{user} is the binary cross-entropy between O_{user} and the label of user identity. Note that the model is trained with the data samples from both real humans and the VAE model. We annotate the identity of the samples from the VAE model with a novel label. This is because we train the VAE model with mixed fall samples from multiple users. The identity is already diluted in the model parameters as well as in the generated samples. As an intuitive example, if the identity is defined as the height of users, the mixed height of two users with heights of 180 cm and 170 cm will create a user with a height of 175 cm. Hence, the generated user can be treated as a novel user. By training the feature extraction, fall detection, and user discrimination networks simultaneously, FallDar successfully fraud the network to misidentify the users and obtain the user-independent latent features for robust fall detection in facing user diversity.

In FallDar, the adopted VAE and adversarial learning technique are commonly used in literature. However, the contribution of adapting them in FallDar is non-trivial. For the data augmentation part, we pinpoint the latent structure (*Kernel Speed*) that underpin the motion diversity of human falls. Based on this, we exploit the feasibility of capturing this motion diversity with VAE models and simulate fall patterns for system training. Compared with existing fall detection works, we are the first to fully analyze and utilize the motion diversity of falls, setting a new stage for human fall detection. For the fall detection part, the key insight is that we can remove human identity from the falling patterns to embrace user diversity. By incorporating a user identification network into the fall detection network, we can remove user identity from the raw signals and force the fall detection network to be resilient to user identity. To the best of our knowledge, we are the first to consider and tackle the user diversity problem in fall detection systems.

7 Evaluation

7.1 Evaluation Setup

Experimental Settings. We collaborate with a leading smart hardware manufacturing company from China to provide fall detection service on their commercial Wi-Fi routers. The ultimate goal of FallDar is to provide a low-cost, sensorless, and contactless fall monitoring system to the elderly that live on their own at home. The prototype system is implemented by MATLAB and Keras [27] mainly for benchmark analysis[1]. The experiments are conducted in the home of a voluntary family and an office room. For the home environment, we managed to collect a large scale dataset when the family members stay at home during the COVID-19 pandemic (the members were all in good health condition). The

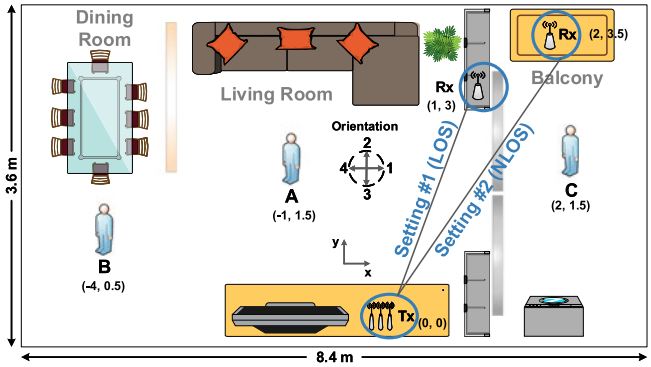


Fig. 17. Experimental layout in a home.

home is a typical household environment that is furnished with desks, sofa, television, cabinet, etc. Fig. 17 sketches the room layout. The size of the monitoring environment is $8.4m \times 3.6m$ and two settings of device deployment are adopted. In setting #2, the LOS path between the transmitter and the receiver is occluded by the door, which is commonly seen in typical home scenarios. The three locations marked with A, B, and C are the places where we collect the fall data. We also mark the four falling orientations with numbers 1 to 4. If we set up a coordinate system with the Tx as the origin and orientation 1 and 2 as x and y axes, the location of Rx and fall are marked in the layout. The Tx is placed at a height of 30 cm, and the Rx in settings #1 and #2 are placed at 2 m and 1.5 m heights respectively. The transmitter and receiver are both equipped with three omnidirectional antennas. We only use one antenna on the receiver device. The Network Interface Cards (NICs) are set to operate on channel 165 with a center frequency of 5.825 GHz and report CSI at a frequency of 1,000 Hz. To synchronize between these devices, a router is adopted to connect them with a server to store the captured data. The office room is furnished with a table and several chairs. The room has a size of around $5m \times 7m$. Fig. 18 sketches the office layout. We place the transmitter on the table and the receiver at the 16 different locations marked with green circles in the layout. For each receiver location, we let the volunteers fall at the four different locations marked with blue squares in the layout.

Data Collection. For the experiment in the home environment, the deployed system continuously captures human activities for six months. Several surveillance cameras are also deployed to help annotate the data instances. To preserve the privacy of the participants, we only perform experiments in the living room, dining room, and balcony. The voluntary family has five members, including two elderly members (the mother and father) and three young members (the son, cousin, and daughter-in-law). Their ages range from 20 to 50. Their heights are within 160 cm to 181 cm and weights are within 50 kg to 75 kg. During six months of monitoring, we captured massive data instances of various human activities. With careful data sanitization and annotation, we select data instances corresponding to only a single participants appear in the monitoring area (the area with grey color in Fig. 17). However, fall activities are rarely seen during data collection. To obtain fall instances, we require the participants to perform fall motions deliberately with controlled risk. Specifically, the participants are

1. All the code and datasets will be released upon publication.

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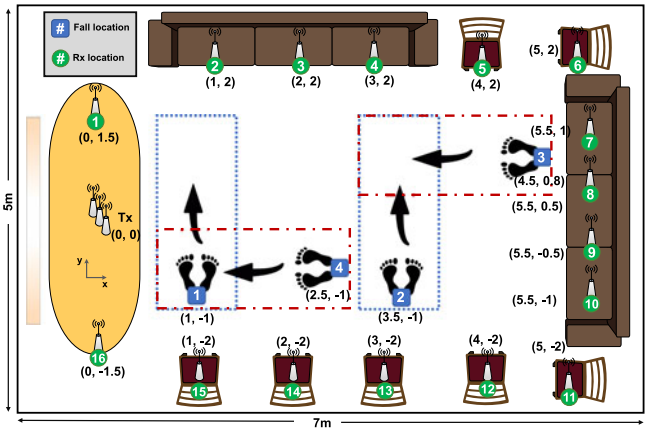


Fig. 18. Experimental layout in an office.

required to wear protective equipment and pretend to fall unconsciously as best as they could. The floor is covered with a sponge to avoid injuries. For the experiment in the office environment, we collected both falling and walking activities from two volunteers. We cover the ground with several mats and let the volunteers fall forward onto them. One volunteer falls at the locations of #1 to #4 and the other one falls at the locations of #1 to #3. We also let one of the volunteers walk freely in the room to collect normal activities. After collecting all the normal and fall instances, we classify the normal activities into 21 classes and fall activities into 6 classes with 24 sub-variations, which are listed in Table 1. We summarize the overview of our dataset in Table 3. When collecting normal data, the users are asked to freely perform these movements in the monitoring area. We have 610 fall instances and 2,000 normal instances collected from the home, among which 115 fall instances are from the participants and the rest are from the dummy. We also have 228 fall instances and 504 normal instances collected from the office room. Besides, we generate 900 instances of synthetic fall instances with our generative model to train the system. All the experiments are approved by our IRB.

Data Segmentation. To build a real-time fall detection system, we design and implement a segmentation algorithm to deal with signal flows. Specifically, we apply a window with a length of 1,5000 ms on the signal streams and slide the window with a step length of 500 ms. In each window, we extract the speed sequence as the input of the detection network. During system training, we design another segmentation algorithm to adjust the window position of fall samples. This algorithm is based on two observations we made in Section 2. (1) The duration of fall is around 1,000 ms to 1,500 ms, in which the falling phase lasts longer than the striking phase. (2) The peak speed is around 3 m/s to 4.5 m/s for typical fall motions. Therefore, for each fall sample, we pinpoint the location of the peak speed and adjust the window to have 1,000 ms on the left and 500 ms on the right of the peak.

TABLE 1
The Observed Fall and Normal Activities

Movements	Types	Sub-variations
Fall	trip, slip, lose-balance, kneel-then-fall, sit-then-fall, walk-then-fall	forward, backward, lateral, on-position
Normal	walking, jogging, sit-down/stand-up, bend-and-pickup, open/close door, open/close fanner, squat, dancing, eating, reading, watching TV, chatting, answering phone, listening to music, laughing, hanging clothes, open/close window	

TABLE 2
The Open Dataset From [12]

Room[1]	A	B	C	D	E	Total[2]
Fall	70	30	2	30	49	181
Normal	67	87	151	112	34	297

1*A: bathroom; B: bedroom; C: corridor; D: kitchen; E: laboratory.*
2*We only use part of the dataset with clear annotation.*

Open Dataset. To fully evaluate the system performance and robustness, we adopt an open dataset released by the authors of [12]. Due to the lack of descriptions, we only adopt the data samples with clear annotations. Table 2 shows the details of this dataset. A total of 181 fall activities and 297 normal activities are collected from five rooms including the bathroom, bedroom, corridor, kitchen, and laboratory.

Metrics. In FallDar, we use two straightforward metrics of fall detection: False Alarm Rate (FAR) and Missed Alarm Rate (MAR). FAR is the ratio between the number of misidentified normal instances and the number of all normal instances, showing how often the users are bothered when no fall event occurs. MAR is the ratio between the number of misidentified fall instances and the number of all fall instances, showing the sensitivity of FallDar to fall events.

7.2 Method Comparison

In this section, we compare FallDar with the state-of-the-art fall detection works based on Wi-Fi signals. We also evaluate the fall detection capability of FallDar with different input fall features and learning methods.

Comparison With the State-of-the-Art Works. We compare FallDar against two state-of-the-art fall detection works, FallDeFi [12] and DeFall [13]. FallDeFi leverages DFS profiles extracted from CSI streams and select several statistical features of them for detection. Then, it uses an SVM classifier for binary classification between fall and normal. DeFall is the first work that proposes to use the speed of body as the features for fall detection. It extracts the speed from the raw CSI streams with an algorithm similar to FallDar and constructs a fall template with the speed and acceleration information. The fall events are identified with a template matching method. To compare FallDeFi and DeFall with FallDar, we split the full datasets (containing 610 real fall instances, 900 synthetic fall instances, and 2,000 normal instances) into train and test and extract the corresponding DFS and speed sequences. We adopt a ten-fold cross-validation method for evaluation. Fig. 19 shows the results. *Intra-domain* represents that the training and testing sets contain the data samples of the same fall types from the same person and the same environment. *Cross-env*, *Cross-user*, *Cross-fall* represent that the training and testing sets contain the data from different environments, different persons, and different fall types respectively. For *Intra-domain* and *Cross-env* cases, FallDar and DeFall achieve better performance

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TABLE 3
Dataset Overview

#	Room	Room Size	Duration	People Number	Samples (fall, normal)	Fall Types	Normal Types
1	Living Room	$4.2m \times 3.6m$	$\sim$ 700 hours	5	1,540 (540, 1000) (495 fall from dummy)	24	21
2	Dining Room	$2.4m \times 3.6m$	$\sim$ 400 hours	5	870 (70, 800)	8	10
3	Balcony	$1.8m \times 3.6m$	$\sim$ 200 hours	5	200 (0, 200)	0	4
4	Office	$5m \times 7m$	$\sim$ 57 minutes	2	732 (228, 504)	1	1

among the three works, which is benefit from the speed features. When testing on different users (Cross-user) and fall types (Cross-fall), FallDar achieves the best performance, demonstrating its robustness to user diversity and motion diversity. From the application perspective, we have to admit that even though FallDar achieves better performance than the existing fall detection systems, the performance is still to be improved for some application scenarios. We give three suggestions to improve the usability of FallDar. First, when FallDar detects fall motions, the system interface can seek for user's confirmation before sounding the alarm for the incident. This can ease the false alarm to some extent. Second, deploying more Wi-Fi devices to increase the signal coverage is another promising method. Last, as we discussed in Section 5, human falls are complex motions with endless types. It will be beneficial to train FallDar with more data samples from both the proposed VAE model and real-world collection. With the above improvements, we believe FallDar will be taken closer to commercial use.

Comparison of Fall Features. We compare three types of different signal features used for fall detection: denoised CSI phase, DFS profile, and speed sequence. For each type of the features, we build a classifier that is similar to the DNN classifier used in FallDar in Fig. 16. Specifically, we adjust the shape of the input layer and keep the other layers unmodified. The denoised CSI phase is extracted from the raw CSI with the denoising algorithms discussed in [16], [17] and PCA is applied to refine the 30 subcarriers into one dimension. Each CSI phase instance has a size of 1,500 (samples during 1.5 seconds)$\times$1 (one principal component). The DFS profile is extracted with the algorithms introduced in [28]. Each DFS instance has a size of 1,500 (samples during 1.5 seconds)$\times$61 (frequency bins). The training and testing sets are both composed of instances from the two environment settings. The evaluation results are shown in Fig. 20. As can be seen, the speed sequence outperforms both the CSI phase and DFS with an average decrease of 35% and 6.6% in terms of FAR+MAR. This demonstrates that the speed sequence is an ideal feature for fall detection.

Comparison of Learning Models. We evaluate the effectiveness of the proposed DNN-based classifier. With speed sequences as input, we classify fall and normal with Naive Bayes, SVM, and the FallDar's classifier. Both training and testing sets contain the instances from all five participants. As is shown in Fig. 21, FallDar's DNN classifier outperforms the other models with an average decrease of 10.9% and 5.9% in terms of FAR+MAR. It is worth noting that even FAR of the SVM model is lower than FallDar, but its MAR is much higher, making it less applicable. The results demonstrate the superiority of the proposed DNN classifier.

Performance on Different Datasets. We evaluate the performance of FallDar on an external open dataset. We refer to our dataset collected in the home environment as #1 and the external dataset as #2. In this experiment, we use four settings. In the first setting, we split dataset #1 into train and test with a ratio of 8:2, each of which contains the samples from all of the users. In the second setting, we mix datasets #1 and #2 and split them into train and test with the same ratio. In the third setting, we train our system with dataset #1 and test it with dataset #2. In the last setting, we only use dataset #2 for training and testing. Note that users' identity is not annotated in dataset #2, and we treat them as from the same user in the second and third settings. Ten-fold cross-validation is used.

The results can be found in Fig. 22. As is shown, the system performance decreases by around 4% for MAR and FAR when trained and tested with dataset #2. The system performance further decrease by around 7% for MAR and FAR when trained and tested with different datasets. The performance decreases are likely caused by the lack of identity annotation in dataset #2. Without annotating the users' identities, the proposed user discriminator network will not work, making the system prone to user diversity. This problem becomes even worse when the training and testing users are different. This reveals that users' identity is a nontrivial factor for fall detection.

Comparison With Outlier Detection Methods. Fall events are rarely seen during daily lives and fall motions induce stronger signal fluctuations than normal motions. Given this fact,

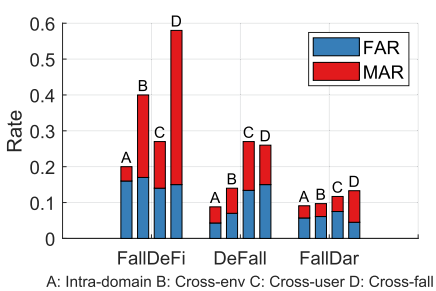


Fig. 19. Comparison between state-of-the-art works.

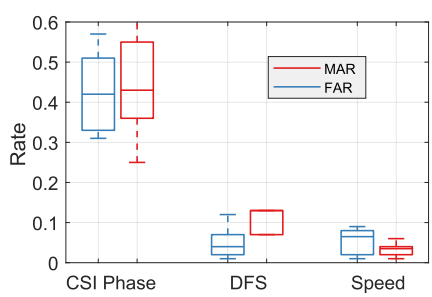


Fig. 20. Comparison between different fall features.

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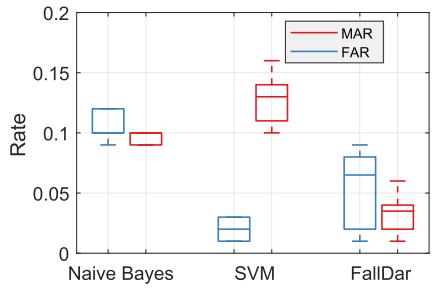


Fig. 21. Comparison between different learning methods.

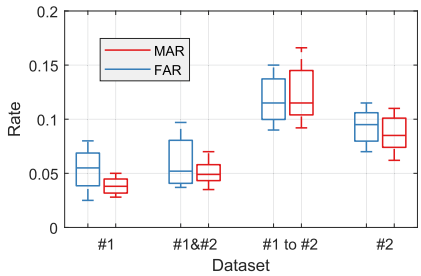


Fig. 22. Performance on different datasets.

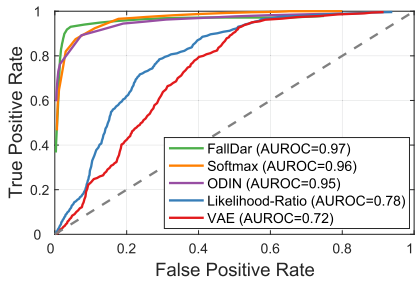


Fig. 23. Comparison with outlier detection methods.

TABLE 4
Outlier Detection Performance

Methods	AUROC↑	AUPRC↑	FPR80↓
FallDar	0.97 (1e-5)	0.94 (1e-4)	0.01 (1e-6)
Softmax	0.96 (6e-5)	0.93 (4e-4)	0.01 (5e-6)
ODIN	0.95 (2e-5)	0.93 (3e-4)	0.01 (2e-6)
VAE	0.78 (9e-5)	0.41 (3e-4)	0.30 (1e-3)
Likelihood-ratio	0.72 (9e-5)	0.30 (5e-4)	0.40 (1e-3)

it is straightforward to model fall detection as an outlier detection problem (while in FallDar we model it as a classification problem). In this experiment, we compare FallDar with four classical outlier detection methods. Note that FallDar is a DNN driven system and we only consider those DNN-based outlier detection methods. 1) The Softmax [29] method recognizes the samples that have low prediction confidence on the Softmax layer as outliers. 2) The ODIN [30] is a variant of the Softmax method. It uses temperature scaling and input preprocessing techniques to detect outliers based on the prediction confidence of the Softmax layer. 3) The VAE method [31] trains a VAE model on resident samples and employs the reconstruction loss during testing as outlier indicator. 4) The likelihood-ratio method [32] is a variant of the VAE method. It contrasts the likelihood of a generative model against a background model and employs the likelihood ratio between the two models as an outlier indicator. The designed fall detection network in Fig. 16 uses the Softmax layer to classify fall and normal instances, hence methods 1) and 2) require no modification to the network. We train the model with normal samples from the users and fall samples from the dummy. For methods 3) and 4), we build a VAE model with the same structure as that we use for data augmentation (Fig. 15). Different from the data augmentation procedure (Section 5.2) that train the VAE with fall samples, we only use normal samples to train the VAE model in this experiment and regard the fall samples as outliers. For method 4), we perturb the training set by flipping the speed sequence, randomly shifting it along the time axis, and scaling the value with a random variable within 0 and 1. The perturbed dataset is used to train the background model. Three metrics are used to depict the outlier detection performance [32], which are the area under the ROC curve (AUROC), the area under the precision-recall curve (AUPRC), and the false positive rate at 80% true positive rate (FPR80). The evaluation results are shown in Fig. 23 and Table 4. Numbers in front and inside of the brackets are mean and variance respectively based on 10 independent runs with random initialization of network parameters and random shuffling of training inputs. As can be seen, the VAE and its variant likelihood-ratio method perform worse than the others. This is because the number of NORMAL motions is tremendous and it is difficult to capture the underlying structures with a VAE model. Consequently, the reconstruction loss of the VAE model is large for both the fall and normal samples. On the contrary, human fall motions have their intrinsic structure (i.e., the Kernel Pattern discussed in Section 5.1), which makes it possible to model human falls and generate synthetic samples with the VAE model. The Softmax and its variant ODIN methods have a decrease in performance compared to FallDar. This reveals that it is crucial to encompass the diversity of human motions when building a fall detection system. Simply treating fall motion as outliers without considering their unique patterns will result in suboptimal performance. While in FallDar, we dive deeper into the fall patterns by modeling it with the speed transformation of the torso. With diverse falling samples as the training dataset, FallDar demonstrates significantly robustness for practical use.

7.3 Robustness Evaluation

Performance on the Speed Estimation Algorithm. In FallDar, we characterize human fall motions with the speed of the torso to have better modeling of the human motion. One of the crucial parts of FallDar to achieve this purpose is the speed estimation algorithm discussed in Section 4. The accuracy of the speed estimation algorithm is important for the overall performance of FallDar. In this experiment, we evaluate the proposed speed estimation algorithm.

We deploy the Wi-Fi transmitter and receiver in both LOS and NLOS settings in the sitting room and the balcony (illustrated in Fig. 17). We invite a user to perform some normal activities and fall activities in location A in the living room. We use a Kinect sensor to capture the motion of the user and log out the speed of the Spine Base joint (defined in Kinect SDK) as the ground truth. We also extract the speed of the human torso with the proposed speed estimation algorithm.

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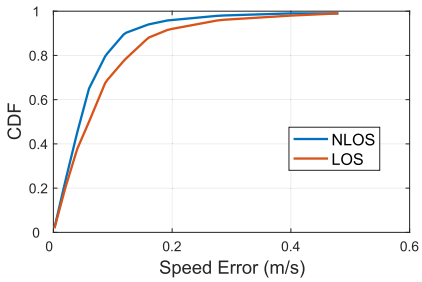


Fig. 24. Performance of the speed estimation algorithm.

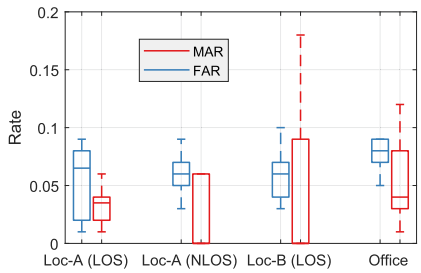


Fig. 25. Performance between different environments.

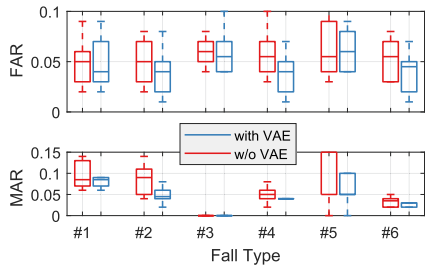


Fig. 26. Performance across fall types.

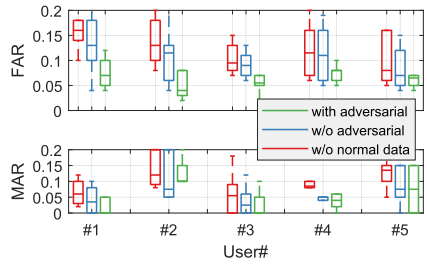


Fig. 27. Performance across users.

The Wi-Fi devices and the Kinect are synchronized with a python script that communicates with socket connection.

Fig. 24 plots the Cumulative Distribution Function (CDF) of the speed estimation errors. A median error of 0.12 m/s and a 90%-tile error of 0.35 m/s in the NLOS case can be observed. The performance decreases slightly for the LOS case, which is because the multipath condition is less obvious in such a case, and the speed estimation model should be based on a rich-scattering environment.

Performance Between Environments. In this experiment, we evaluate the performance of FallDar when it is trained and tested with data from different environments. As is illustrated in Fig. 17, the Wi-Fi devices are placed in two different settings. In Setting #1, both transmitter and receiver are placed in the living room and have LOS between them. In Setting #2, the transmitter is placed in the living room and the receiver is placed in the balcony and they are occluded by the door. In this experiment, both the training dataset and the test dataset are composed of the fall data from the dummy and the normal data from the five participants. The training datasets are collected in the Dining Room for the LOS setting. The 900 instances of synthetic fall data are also used for training. This experiment contains four settings. In the first setting, we evaluate the performance when test data are collected in the same location and setting as that in the training set. In the second setting, we evaluate the performance when test data are collected in the Dining Room in the NLOS setting. In the third setting, we evaluate the performance when test data are collected in the living room in the LOS setting. In the fourth setting, we evaluate the performance with the data collected from the office environment. The results are shown in Fig. 25. As can be seen, FAR and MAR are both consistent when the location and LOS condition changes. The performance also stays consistent across different rooms. Benefiting from the environment-independent attribute of the extracted speed sequence, FallDar is robust to environmental diversity.

Performance Between Fall Types. In this experiment, we evaluate the capability of FallDar when dealing with diverse fall types. We consider a total number of 6 fall types and 24 of its sub-variations in FallDar, including trip, slip, lose-balance, kneel-then-fall, walk-then-fall, and sit-then-fall, which are labeled from #1 to #6. These fall data are from the dummy and the five participants and the normal data are collected from the five participants. We train the model with the data from five out of the six fall types and part of the normal types. We compare the performance when the 900 instances of synthetic fall data are used or not used for training. The data of the last fall type and the remaining normal data are used for testing. The results are shown in Fig. 26. As can be seen, with the synthetic fall data used for training, FallDar can decrease its fall detection accuracy by 0.78% for FAR and 3.4% for MAR when novel fall types occur. This result demonstrates that augmenting the training datasets with the generated fall data can enhance the system's robustness to motion diversity.

Performance Between Users. Existing fall detection works are trained and tested on the same users, while FallDar proposes a novel classifier based on adversarial learning to enhance the system's robustness to user diversity. In this experiment, we evaluate the performance when the system is tested on new users with no fall data collected from them for training. The training set is composed of the fall data from the dummy and the normal data from all the participants (including the new users). The 900 instances of synthetic fall data are also used for training. For testing datasets, we collect 20 fall instances and 100 normal instances from each of the five participants. We evaluate the performance with and without the adversarial part of the classifier. We also evaluate the performance when removing the normal data of the test user from the training set. The results are shown in Fig. 27. As can be seen, FAR and MAR decrease by about 4.5% and 0.8% respectively when the adversarial learning method is adopted. This demonstrates the effectiveness of the adversarial method to deal with user diversity. Besides, the FAR and MAR decrease by about 6.1% and 3.2% when removing the normal data of the test user from the training set. This confirms that by using the

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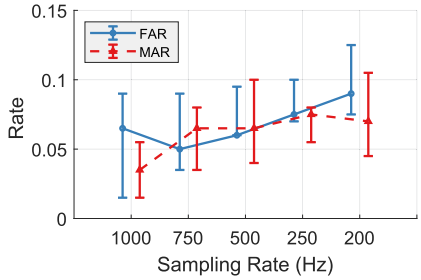


Fig. 28. Impact of CSI sampling rate.

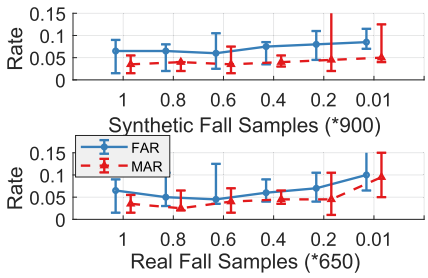


Fig. 29. Impact of training samples.

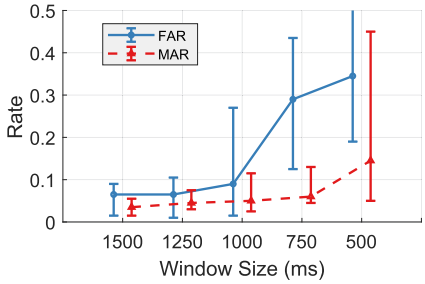


Fig. 30. Impact of window size.

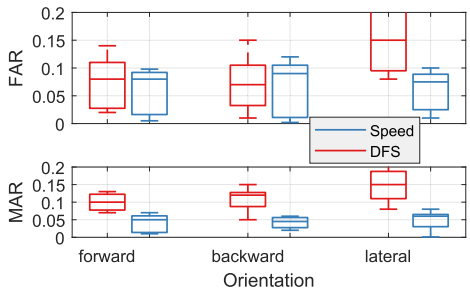


Fig. 31. Impact of falling orientations.

normal data samples for adversarial training, system performance can be more robust to the user diversity problem.

Performance Between Fall Orientations. The proposed speed feature is theoretically robust to moving orientations. In this experiment, we evaluate the performance when the users fall at different orientations. The training data are collected from five participants and the dummy in the home environment. The synthetic fall samples are also used for training. We use the data from one of the falling orientations for testing and the others for training. The performance of DFS and speed features are compared. Fig. 31 shows the evaluation results. As can be seen, the performances of the speed feature for different orientations are consistent. On the contrary, the performance of DFS is susceptible to orientation changes. This is because DFS only captures the radial velocity, and falling in the tangential direction will induce minor doppler frequency components. This confirms our observations in Figs. 11 and 12.

7.4 Parameter Study

Impact of CSI Sampling Rate. In the above experiments, we evaluate FallDar with a CSI sampling rate of 1000 Hz. However, this rate is not guaranteed as the Wi-Fi packet transmission might be interfered. In this experiment, we evaluate the impact of CSI sampling rate on system performance. We collect CSI at a rate of 1000 Hz and downsample the CSI series to 750 Hz, 500 Hz, 250 Hz, and 200 Hz for fall detection. The input shape of the DNN models is adjusted accordingly. The results are shown in Fig. 28. As can be seen, FAR and MAR drop by 3.6% and 4.1% respectively when the sampling rate decreases from 1000 Hz to 200 Hz. This is because fall movements are featured by the proliferation of speed with a maximum of about 4-5 meters/second, which could hardly be captured by a low sampling rate. In practice, FallDar employs shorter Wi-Fi packets to mitigate the impact on communication.

Impact of Training Samples. In this experiment, we evaluate the impact of the training samples on detection accuracy. The total datasets are composed of the 610 real fall instances, the 900 synthetic fall instances, and the 2,000 normal samples. In the first case, we gradually decrease the number of synthetic fall samples from the datasets and perform ten-fold cross-validation. In the second case, we gradually decrease the number of real fall samples from the datasets and perform ten-fold cross-validation. The results are shown in Fig. 29. As can be seen, when decreasing the number of synthetic fall samples from 900 to 10, FAR and MAR drop by 1.7% and 1.9%. When decreasing the number of real fall samples from 610 to 5, FAR and MAR drop by 7.2% and 6%. This demonstrates that through the collaborative use of real fall data and the synthetic fall data, FallDar can boost its performance to a significant extent.

Impact of Window Size. In the above experiments, each instance in the datasets corresponds to a time window of 1.5 seconds. This is based on our empirical observation that all fall movements occur within 1.5 seconds. In this experiment, we evaluate the impact of the window size on the detection accuracy. We randomly crop the data into a time window of $1,250ms$, $1,000ms$, $750ms$, and $500ms$. The results are shown in Fig. 30. As can be seen, FAR and MAR slightly drops by 3.2% and 1.7% when the window size decreases from $1,500ms$ to $1,000ms$. But when window size further decreases from $1,000ms$ to $500ms$, FAR and MAR tumble by 30% and 17%. This demonstrates that a time window of $1,000ms$ suffices for fall detection. In practice, we employ $1,500ms$ for better performance.

8 Related Work

RSSI-Based Activity and Fall Recognition. RSSI measurements from commodity Wi-Fi devices have been leveraged to recognize gestures [33] and many kinds of human activities (including falls events) [34], [35]. WiGest [33] leverages the effect of in-air hand motion on RSSI to recognize gestures. It divide the impact signals into three states and uses the

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combinations of the states to perform gesture recognition. Harmony [34] transforms RSSI into frequency component vectors and uses machine learning technique to classify nine kinds of human activities. [35] uses software radio to improve the granularity of RSSI values and consequently improve the accuracy of activity recognition. However, the above RSSI-based methods have low resolution and have intrinsic limitations on the improvement of recognition accuracy [36], [37].

CSI-Based Activity Recognition. CSI is commonly used to recognize activities. E-eyes [38] collects location-oriented time-domain channel features to distinguish in-place activities. CARM [28] employs a CSI-speed model to build correlations between time-frequency features and specific activities. On the basis of CARM, WiFiU [24] generates limb and torso velocities in CSI for gait recognition. WiDance [16] extracts motion-induced DFS to recognize human motions. In addition, DeMan [39] uses amplitude and phase information of CSI to detect the moving and stationary users and WiDir [40] uses phase change of CSI to estimate user's walking direction. WiTrack2.0 [41] and Widar2.0 [42] enable human localization with wireless signals. [43], [44], [45] extract Doppler shifts or higher level features from wireless signals to perform human gesture recognition. [20], [46] enables human identification with Wi-Fi signals based on gait patterns. These systems are pioneers of exploration in CSI analysis and inspire the idea of Wi-Fi-based fall detection.

CSI-Based Fall Detection. Among the CSI-based fall detection systems, WiFall [10] uses fine-grained CSI and a one-class Support Vector Machine classifier and Random Forest algorithm to classify different human activities and then identify falls events. RT-Fall [11] exploits time-frequency domain patterns of CSI to automatically detect falls events in real-time. FallDeFi [12] also applies the time-frequency domain features and design a sequential forward selection algorithm to selects specific features that are resilient to environmental changes. TL-Fall [47] further solves the issue that a fall detection model trained in one specific environment does not work well in the other. It leverages transfer learning theory to further reduce the environmental impact on fall detection. FallViewer [48] proposes a series of calibration algorithms to calibrate the raw Wi-Fi signals that are influenced by the multipath effects and mitigate the influence on the environment. Comparing to these works, FallDar aims to be resistant to environmental diversity, motion diversity, and user diversity simultaneously, which has prominent advantages among the state-of-art works and promotes the commercialization and productization of fall detection in smart Wi-Fi devices. DeFall [13] is a pioneering work to extract environment-independent speed features for fall detection. Our speed estimation is similar to that in DeFall. However, the recognition algorithm in that paper is based on template matching and is constrained to simple fall motions, which necessitates the design of data augmentation and user discrimination algorithms in FallDar.

Other Methods of Fall Detection. There exists other means focusing on fall detection including vision-based, IMUs-based, and ambient device-based methods. Vision-based methods analyze images recorded by depth cameras, such as Kinect, to monitor falls events [2], [3], [4], [49]. IMUs-based methods usually leverage accelerometers, gyroscopes, and inclinometers to recognize falls events from signal changes [5], [6], [7], [50]. Among the IMU-based fall detection works, [51], [52] present some deep learning-based methods to augment the fall dataset, which could be valuable for Wi-Fi-based fall data augmentation. In terms of ambient device-based methods, G-Fall [53] uses geophones to decompose falling modes and acoustic-FADE [54] uses a microphone array to identify if falls events happen. Different from these works, FallDar enables a secure, device-free, and ubiquitous fall detection system.

9 Limitations and Discussions

FallDar pushes forward Wi-Fi-based fall detection in three aspects including user diversity, motion diversity, and environment diversity. Even though FallDar achieves better performance in the above three aspects, there are still some limitations that demand to be resolved.

The Impact of Body Shapes on Residual Speed. The protective movements of human limbs can perturb the falling pattern, which will deteriorate the recognition performance. To alleviate this effect, FallDar proposes to disentangle the residual speed from the falling speed pattern to characterize the motion diversity and use it for motion simulation. The residual speed is obtained by subtracting the speed pattern of a dummy from that of the real users. However, using the dummy is not the optimal way because the dummy has different body shapes from that of the real users. The residual speeds contain undesired information corresponding to the body shape discrepancies between the dummy and the real users. To tackle this problem, one potential solution is to build custom dummies that have similar body shapes to that of the real users. This seems cumbersome but could be effective. Other solutions are expected in future works.

The Impact of Data Imbalance on User Identification. The discrepancies between different users can have an impact on fall detection performance. To resolve this problem, FallDar incorporates a user identification module into the fall detection network. By deliberately forcing the network to misidentify users, we train the network to discard the user identities. However, the training datasets are imbalanced. The datasets collected from the dummy and generated by the simulation module only contain falling samples, and most of the real users only have normal data samples. The different number of falling and normal data samples from different sources could be harmful to the user identification algorithm. To remedy this problem, some potential solutions include collecting the falling samples from all the real users and simulating some normal data samples.

Limitations in Dataset Scale. In the experiments, we collect data from three indoor environments and five volunteers. The scale of the dataset is sufficient for benchmark analysis but still not enough for deployment. In real-world applications, the number of environments and users could be enormous. To improve the application prospects, other data augmentation solutions could be valuable besides the designed VAE-based method. For example, some works [55], [56] have demonstrated the feasibility of simulating RF signal measurements from three-dimensional models. Therefore, we can build models of more kinds of indoor environments and users, and use them to simulate more

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data samples. This simulation process does not require the users to perform falling movements and can be easily adapted to more environments and users.

High False Alarm Rate. FallDar has an overall false alarm rate of 5.7% and a missed alarm rate of 3.4%. The performance is better than the state-of-the-art works but still has a gap from the real-world applications. To further reduce the false alarm rate, one promising solution is to consider body height. After falling, the human body will be close to the ground, having a lower height than the other normal activities, which may also induce high speed. By considering the body height, some normal but abrupt motions will be filtered. Another potential solution is to use the body skeletons as features rather than the speed. Some works [57], [58] have demonstrated the feasibility of reconstructing the human skeleton from RF signals. The human skeleton contains richer information regarding human falling status compared to speed. By exploiting skeleton for fall detection, some fall-like movements will be filtered out, reducing the false alarm rate to some extent.

10 Conclusion

In this paper, we propose a Wi-Fi-based passive fall detection system. To enhance the system's robustness to environmental diversity, we design a novel speed extraction algorithm to extract the body speed as a fall feature. To enable FallDar to detect more types of fall, we design a novel generative DNN model, which leverages a small dataset of real data and can generate enormous synthetic datasets for training. To enhance the system's robustness to user diversity, we design a novel adversarial learning-based DNN classifier that can extract user-independent fall features. Through the collaborative working of the above methods, FallDar outperforms the state-of-the-art fall detection works.

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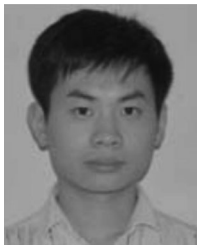
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